PROJECT 2: Employee Performance & Salary Analysis

We'll follow this structure:

1. Load Data
2. Understand Dataset
3. Data Cleaning
4. Salary Analysis (NumPy)
5. Performance Analysis (Pandas)
6. Visualization (Matplotlib)
7. Business Insights
8. Resume Description

Step 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Step 2: Load Dataset

Put CSV file inside your Jupyter folder.

```
df = pd.read_csv("HR-Employee_data.csv", encoding="latin1")
df.info()
```

What is latin1 encoding?
latin1 (also called **ISO-8859-1**) is a **single-byte character encoding**.

👉 **Key property:**

- It can decode **any byte value from 0–255** without throwing an error.
- That's why it's often used as a **safe fallback encoding**.

❌ **Why UTF-8 sometimes fails**

When you read a file like this:

pd.read_csv("data.csv")

- Python assumes **UTF-8** by default.
- **If the file contains special characters (like smart quotes, accents, symbols), you may see:**

UnicodeDecodeError: 'utf-8' codec can't decode byte 0x92

This happens because

- The file was saved using **Windows / legacy encoding**
- UTF-8 cannot interpret certain byte patterns

✅ **Why encoding="latin1" works**

Because **latin1 never fails decoding**, this works:

pd.read_csv("data.csv", encoding="latin1")

- ✓ No UnicodeDecodeError
- ✓ File loads successfully
- ✓ All bytes are mapped to characters

STEP 3: Understand Data

```
df.shape                    # rows & columns   i.e (1470, 35)
df.columns                  # column names
df.info()                   # data types
df.describe()               # statistical summary
```

Note: Important columns for your project:

- MonthlyIncome
- PerformanceRating
- Department
- JobRole
- YearsAtCompany
- Attrition *i.e* means when an employee leaves the company

Your project goal: ***Analyze salary distribution and performance ratings***

So let's understand why each column is important.

1️⃣ MonthlyIncome 💰 (Core Target Variable)

Why important?

➔ This is the **main variable** of your project.

You are analyzing:
- Salary distribution
- Mean / Median / Percentiles
- Salary by department
- Salary vs performance
- Salary vs attrition

Without this column → No salary analysis.

Business Meaning:

Helps HR understand:
- Pay inequality
- Salary gaps
- Compensation strategy

2️⃣ PerformanceRating ⭐

Why important?

➔ Because your project includes: ***Performance & Salary Analysis***

You need to check:
- Do high performers earn more?
- Is salary justified by performance?
- Is performance linked to attrition?

Example analysis: `df.groupby("PerformanceRating")["MonthlyIncome"].mean()`

Business Insight:

If high performers are not paid more → Risk of losing talent.

3️⃣ Department 🏢

Why important?

➔ Different departments earn differently.
Example:
- R&D
- Sales
- HR

You analyze: `df.groupby("Department")["MonthlyIncome"].mean()`

Business Insight:

- Which department has highest salary?
- Is Sales paid more than HR?
- Are some departments underpaid?

4️⃣ JobRole 👔

Why important?

➔ More detailed than Department.

Example:

- Manager
- Sales Executive
- Research Scientist

This helps analyze:

- Salary by role
- Role-wise performance
- Role-wise attrition

Note:- *This makes your project deeper and Interviewers love this level of breakdown.*

5️⃣ YearsAtCompany ⏳

Why important?

To analyze:

- Experience vs Salary
- Experience vs Performance
- Experience vs Attrition

Example: `df.groupby("YearsAtCompany")["MonthlyIncome"].mean()`

Business Insight:

- Does salary increase with experience?
- Do new employees leave more?
- Are senior employees underpaid?

This is strategic HR insight.

6️⃣ Attrition 🚪

Why important?

Attrition = Employee Left or Stayed.

Now you can analyze: `df.groupby("Attrition")["MonthlyIncome"].mean()`

This tells:

- Do low salary employees leave more?
- Do high performers leave?
- Which department has highest attrition?

This makes your project **real-world applicable**.

🔥 Why These Columns Together Are Powerful

When combined, you can answer:

- Are high performers paid fairly?
- Is salary linked to attrition?

- Do experienced employees earn more?
- Which department has pay inequality?
- Is company retaining top talent?

This moves your project from:

❌ Just Data Analysis
to
✅ Business Decision Analysis

Summary	
Column	**Why Important**
MonthlyIncome	Core salary variable
PerformanceRating	Check if high performers earn more
Department	Compare salary across departments
JobRole	Role-wise pay structure
YearsAtCompany	Experience vs salary growth
Attrition	Who left and why

🎯 **How to Explain in Interview**

Say this confidently:

"I selected MonthlyIncome as the primary metric and analyzed it across PerformanceRating, Department, JobRole, YearsAtCompany, and Attrition to derive business-level HR insights regarding compensation fairness, pay dispersion, and employee retention."

That sounds like an Analyst 🔥

STEP 4: Salary Analysis (NumPy Practice)

1. **Extract salary column:** salary = df["MonthlyIncome"]
2. **Total:** np.sum(salary) *i.e* np.int64(9559309)
3. **Mean Salary:** np.mean(salary) *i.e* np.float64(6502.931292517007)
4. **Median Salary:** np.median(salary) *i.e* np.float64(4919.0)
5. **25th & 75th Percentile**
   np.percentile(salary, 25) *i.e* np.float64(2911.0)
   np.percentile(salary, 75) *i.e* np.float64(8379.0)

📌 **Interview Tip:**
- Mean affected by outliers
- Median better for salary distribution

WHY?

1. **Mean tells:** 👉 Average salary in company.

   But…
   Mean can be distorted by very high salaries (outliers).

2. **Median shows:** 👉 The middle employee salary.

   It is better for salary data because:
   - Salary usually has extreme high earners.
   - Median is stable.

   If Mean > Median → Data is right-skewed.

3. **Why 25th & 75th Percentile?** They show: 👉 Middle 50% salary range.

   This helps understand:
   - Salary spread
   - Income inequality
   - Typical earning range

Notes

What is Percentile?

A **percentile** tells us: "What value lies below X% of the data?"

✅ 25th Percentile (Q1)

```
np.percentile(salary, 25)
```

This means:
👉 25% of employees earn **less than or equal to** this salary.
👉 75% earn more than this amount.

It is also called **First Quartile (Q1)**.

✅ 75th Percentile (Q3)

```
np.percentile(salary, 75)
```

This means:
👉 75% of employees earn **less than or equal to** this salary.
👉 Only 25% earn more than this.
It is also called **Third Quartile (Q3)**.

📊 Example (Simple Numbers)

Suppose salaries are: 10k, 20k, 30k, 40k, 50k

- 25th percentile ≈ 20k
- 75th percentile ≈ 40k

Meaning:
- Bottom 25% earn around 20k
- Top 25% earn above 40k

🎯 Why 25th & 75th Percentile Matter in Salary Analysis

They help you understand:
- Salary spread
- Income inequality
- Mid-level salary range
- Outliers

The range between Q1 and Q3 is called:

📦 **IQR (Interquartile Range)** IQR = Q3 - Q1

- It shows where the middle 50% of employees lie.
- This is what boxplots display.

🧠 Now Interview Tip Explained

❌ Mean is affected by outliers

Example: 10k, 12k, 15k, 18k, 500k

Mean = very high (because of 500k)
That one high salary increases the average.

✅ Median is better for salary distribution

Median = middle value

For same data: 10k, 12k, 15k, 18k, 500k
Median = 15k
Median ignores extreme values.
👉 That's why in HR / salary analysis, median is more reliable.

📌 Real Insight for Your Project

When you run:

```
np.mean(salary)
np.median(salary)
```

You might see: Mean > Median
That means:
📈 Salary distribution is **right-skewed**
(Few employees earn very high salaries)

🎯 How to Explain in Interview

You can say:

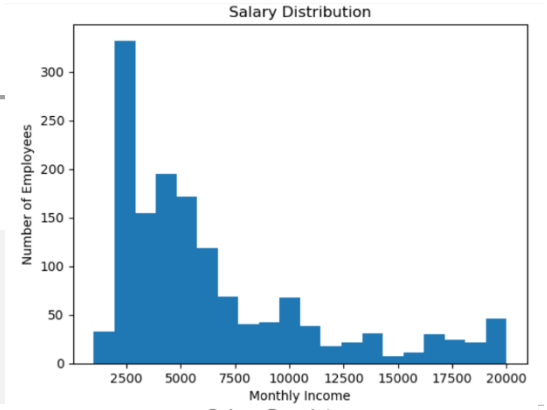
"I used percentiles to understand salary distribution. The 25th and 75th percentiles helped identify the salary range of the middle 50% employees. Since salary data is usually right-skewed, I relied more on median rather than mean because mean is affected by extreme high-income outliers."

That sounds professional 🔥

STEP 5: Salary Distribution Visualization

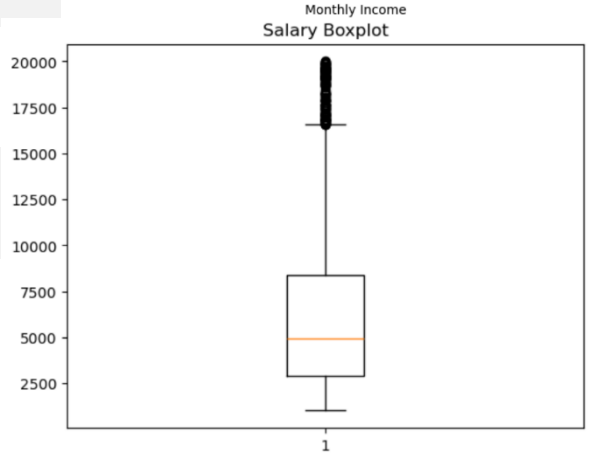
📊 Histogram

```
plt.hist(salary, bins=20)
plt.xlabel("Monthly Income")
plt.ylabel("Number of Employees")
plt.title("Salary Distribution")
plt.show()
```



📦 Boxplot

```
plt.boxplot(salary)
plt.title("Salary Boxplot")
plt.show()
```



Boxplot shows:

- Median
- Outliers
- Salary spread

STEP 6: Performance Analysis

1. Check unique ratings:

```
df["PerformanceRating"].value_counts()
```

Output:
```
PerformanceRating
3    1244
4     226
Name: count, dtype: int64
```

Note: WHY? To compare categories.

Helps answer:

- Which department earns more?
- Are high performers paid more?
- Do low salary employees leave?

Groupby = Business decision tool.

2. Average Salary by Performance

```
df.groupby("PerformanceRating")["MonthlyIncome"].mean()
```

Output:
```
PerformanceRating
3    6537.274116
4    6313.893805
Name: MonthlyIncome, dtype: float64
```

3. Average Salary by Department

```
df.groupby("Department")["MonthlyIncome"].mean().sort_values(ascending=False)
```

Output:
```
Department
Sales                    6959.172646
Human Resources          6654.507937
Research & Development   6281.252862
Name: MonthlyIncome, dtype: float64
```

4. Salary of Employees Who Left (Attrition)

```
df.groupby("Attrition")["MonthlyIncome"].mean()
```

Output:
```
Attrition
No        6832.739659
Yes       4787.092827
Name: MonthlyIncome, dtype: float64
```

Note: WHY?

Because: Companies want to know:

👉 Why employees leave.

- If low-paid employees leave more → salary dissatisfaction.
- If high performers leave → talent retention issue.

This is powerful for HR analysis 🔥

STEP 7: Advanced Analysis (Important for Resume)

High Performers vs Low Performers

```
high = df[df["PerformanceRating"] == 4]
low = df[df["PerformanceRating"] == 3]
```

```
np.mean(high["MonthlyIncome"])
```
Output: np.float64(6313.8938053097345)

```
np.mean(low["MonthlyIncome"])
```
Output: np.float64(6537.2741157556275)

Experience vs Salary

```
df.groupby("YearsAtCompany")["MonthlyIncome"].mean()
```

```
YearsAtCompany
0     4113.500000
1     4681.304094
2     4953.511811
…     ……————————
36    19315.500000
37    13872.000000
40    10312.000000
Name: MonthlyIncome, dtype: float64
```

STEP 8: Business Insights (Very Important)

You must write insights like this:

Example:

- Most employees earn between X and Y range.
- High performers do not earn significantly more than average performers.
- R&D department has highest average salary.
- Employees who left have lower average income.
- Salary distribution is right-skewed.

⚠️ This is what interviewers look for — NOT just code.

✅ **STEP 9: Resume Description (Use This)**

You can write:

Conducted exploratory data analysis on IBM HR dataset (1470 employees) to analyze salary distribution and performance trends.
Applied NumPy for statistical metrics (mean, median, percentiles) and Pandas for group-based analysis.
Visualized salary spread using histograms and boxplots.
Generated HR insights on attrition and performance-based salary gaps.